

Marco Villasuso

<https://marco-villasuso.github.io/>

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Summary

Aerospace engineering undergraduate at the University of Colorado Boulder (expected May 2028) with hands-on experience in propulsion, flight vehicle systems, CAD, FEA, CFD, and hardware integration. Led a team of ~10 people through design and iteration of a scale flight vehicle and supported a ~800 lbf Ethanol/LOX engine hot-fire.

Education

University of Colorado Boulder - BS Aerospace Engineering

August 2024 - May 2028 (Expected)

- Maintained a GPA of 3.56 / 4.0, Dean's List for 3 of 4 semesters
 - Relevant courses: aerodynamics, structures, vehicle dynamics, systems & integration
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Leadership / Projects / Work Experience

Big Dumb Rockets (BDR)

August 2025 - Present

- **Team lead for Hopper Demo.** Led a ~10 person team through systems design and integration developing a scale TVC/GNC hopper demo. Wrote an interface control document (ICD) describing structures, electronics, and telemetry. Conducted trade studies and reviews on motor selection to target a vehicle thrust-to-weight ratio (**TWR**) **near 1.5** and simulate a full-scale hopper.
- Kept hopper demo within an allotted budget of \$800. **Presented preliminary design review (PDR)** to a panel of collegiate engineers and advisors. Manufacturing begins in November.
- **Team lead for Capacitive Fill**, a cryogenic tank fill sensor. Led a 3 person team through design, manufacturing, and integration. Worked with machinists for integration into model tanks. Successfully measured tank fill level from empty to full with **± 5% accuracy** of fill height using distilled water as a low-cost proof of concept.
- Executed pre-fire checkouts, system pressurization, and real-time troubleshooting for Regen Mk I, supporting a successful **~800 lbf hot-fire for 5 seconds**. Diagnosed an incorrectly attached tank pressure transducer and resolved before ignition.
- Performed hardware integration and acceptance testing for Regen Mk I, including component inspection, assembly verification, and post-test teardown analysis. Assisted with both injector manifold machining and LOX-cleaning before integration.
- Evaluated Regen Mk II's thrust plate under 1,500 lbf thrust vectoring loading conditions in SOLIDWORKS, concluding that its factor of safety (**FoS**) **was below 1**, delivered results back to the thrust plate's responsible engineer and suggestions for a redesign.

RC Jet (Solo)

July 2026 - October 2026

- Independently designed, simulated, built, and flew a ducted-fan powered remote controlled jet. Designed in OnShape, ran FEA simulation for wing loading in SOLIDWORKS, and performed flow analysis in flow5, estimating a top speed of ~50 mph. Fabricated via 3D printing through 3 design iterations, improving on wing inner geometry for rigidity. Flew one successful powered flight for 30 seconds, achieving a maximum altitude of 65 feet.

CU Book Store

August 2024 - Present

Student Lead for Technology (December 2024 - Present)

- Managed and supported technical systems including 3D printers, laser cutters, computers, tablets, kiosks, and CAD-related equipment. Performed troubleshooting of both hardware and software issues, and coordinated repairs and deployment of technical equipment.
 - Tested and deployed a campus-wide 200 computer rental-system implementation and trained staff on new equipment and procedures, yielding a **net revenue of > \$80,000** in the first year after launch.
 - Conducted employee training and performance evaluations while coordinating technical tasks across the team, increased team size from 4 to 8.
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Technical Skills

- **CAD:** SOLIDWORKS, OnShape
- **Manufacturing:** 3D printing (PLA, ABS, ASA, SLA), laser cutting, 2-axis CNC, MIG welding
- **FEA:** SOLIDWORKS Simulation
- **CFD:** Ansys Fluent, flow5
- **Systems:** Systems Integration, Flight Hardware, Propulsion Systems, Cryogenic Systems
- **Testing & Operations:** Engine Hot-Fires, Hardware Acceptance Testing, LOX Handling, Troubleshooting
- **Programming:** C & C++, Python, MATLAB, Lua